

Quantum Voting Protocol from Classical Assumptions

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Abstract

Quantum voting allows us to design voting scheme by quantum mechanics. The existing quantum voting protocols mainly use quantum entangled states. However, the existing protocols rarely consider the problem of repeated voting and tampered voting by malicious voters, and hybrid quantum voting protocols have not been discussed. In this paper, we use EFI pairs (Entity-Friendly Integer pairs) instead of quantum entangled states to address the shortage of existing protocols, and propose a new quantum voting protocol. Our protocol is structured to avoid repeated voting by any voter, and can prevent the leakage of voters' voting information. The security of our protocol can be finally reduced to a classical assumption i.e. $BQP = QMA$. Combined with quantum key distribution (QKD), we further optimize the protocol to prevent malicious adversaries from interfering with the final voting results. Moreover, we use extended noisy trapdoor claw-free function (ENTCF) to construct the first hybrid quantum voting protocol, which allows a classical voter to interact with a quantum center through a classical channel to complete the voting process.

Keywords: Quantum Voting, EFI pairs, ENTCF, Hybrid quantum, Quantum Commitment, Complexity Class

1 INTRODUCTION

Voting is widely considered as a crucial and efficient decision-making mechanism. However, classical voting protocols are increasingly susceptible to the quantum attacks, primarily because their security mainly relies on classical hard problems, like factoring large integers which are possibly improved by rapidly-developing quantum techniques [1]. Quantum voting protocol (QVP) is a scheme of applying quantum technology to the voting process. It uses the principles of quantum mechanics, such as quantum no-cloning theorem, to ensure the security, integrity and anonymity of voting information.

Vaccaro, Spring and Cheffes [4] firstly propose QVP in 2007, and propose the constructions about anonymous voting in various scenarios based on the entangled state shared between at least two sites, but these protocols are based on only one variable. Later, Jiang L, He G and Nie D [5] propose QVPs based on continuous variables in 2012, while they only consider one kind of quantum entanglement. In 2016, Wang Q, Yu C and Gao F [6] consider two kinds of entangled quantum states to design a quantum anonymous voting protocol. Combining more tools, Saiyam Sakhuja and S. Balakrishnan [7] design a blending construction applying to quantum entanglement and superposition, block-chain technology and quantum digital signatures in 2024. However, all existing constructions are based on assumptions of quantum mechanics instead of classical assumptions, this leads to the needs of the performance of equipments and communication environment for these protocols, which means they will inevitably lose accuracy when the quantum equipment is damaged for some reasons or the noise of the quantum channel is high. In addition, the existing protocols rarely consider the problem of repeated voting by malicious voters, as a result, the final counted voting results may be manipulated by malicious adversaries. Moreover, all existing protocols assume that both parties have quantum computing capabilities, while the situation of classical voters voting at the quantum center is still an open issue. Therefore, how to use methods other than quantum mechanics to ensure the properties, i.e correctness, fairness and verifiability, of QVP, and how to construct a protocol for hybrid quantum voting are urgent research problems.

In this paper, we propose a new quantum voting protocol based on EFI pairs and the first hybrid quantum voting protocol. The security of our first protocol comes from EFI pairs' computational indistinguishability, and the security of EFI pairs is essentially reduced to a classical assumption $BQP=QMA$ [12] instead of QVP that based on quantum entangled states being reduced to the basic assumptions of quantum mechanics in most cases, and such a classical assumption is a weaker prerequisite. EFI pairs' statistical distinguishability is a basis for counter who holds key to verify the vote. Also, statistical distinguishability of EFI pairs avoid the situation of repeat voting counter. Moreover, we adapt the structure of EFI pairs and associate with the property of quantum key distribution so that our protocol can resist more attack models of malicious adversaries. The security of our second protocol is based on ENTCTF, which can be reduced to the Learning With Errors (LWE) problem. This means that our protocol can resist the attack of malicious adversary. For the first time, this protocol enables classical voters to complete voting only by interacting with the quantum center through the classical channel.

Our contributions can be summarized as follows.

Firstly, we present the construction of a quantum voting protocol that can avoid repeated voting. This advantage enables our protocol to be applicable to a wider range of scenarios and achieve higher security strength. Secondly, we weaken the secure assumption of QVP. Instead of quantum non-cloning theorem and uncertainty principle, the security of our QVP reduce to a classical assumption i.e. $BQP = QMA$. This change makes our protocol more practical than any existing QVP without loss security. Thirdly, we propose the first hybrid quantum voting protocol, which reduces the computational power requirements of quantum voting and greatly reduces the hardware requirements of protocol runtime. We compare our protocol with existing protocols in terms of assumptions, number of rounds and adversary model, and list the results in table 1.

	Assumptions	Number of rounds	Eavesdropper	Dishonest	Repeat	Tamper
[4]	Entanglement	4	✓	×	×	×
[5]	Entanglement	n	✓	×	×	×
[6]	Entanglement	3	✓	✓	×	×
[7]	Entanglement	12	✓	✓	×	×
Our QVP 1	$BQP=QMA$	1	✓	✓	✓	×
Our QVP 2	$BQP=QMA$	Round(QKD) + 1	✓	✓	✓	✓
Our HQVP	LWE	5	✓	✓	✓	✓

Table 1 Comparison with Existing Quantum Voting Protocols

The remainder of this paper is organized as follows. In Section 2, we introduce notations, and the relevant definitions and theorems in quantum cryptography and complexity theory. In Section 3, we construct a protocol based on EFI pairs against semi-honest adversary (honest but curious). In Section 4, we further propose a protocol based on QKD against malicious adversary. In Section 5, we propose the first hybrid quantum voting protocol, and analysis the security of our protocol. We then conclude with some discussions and open problems in Section 6.

2 PRELIMINARIES

Unless otherwise specified, the symbols used in this part apply to the paper. We use $\mathcal{H}$ to represent a finite-dimension Hilbert space, a mathematical space in quantum mechanics used to describe the state of a quantum system, and use $\mathcal{H}_A$ and $\mathcal{H}_B$ to represent the specify distinct spaces. We write $L(\mathcal{H})$ as a set of linear operators on $\mathcal{H}$ used to describe quantum operations, and $S(\mathcal{H})$ as a set of pure quantum states on $\mathcal{H}$. We use $I_A \in L(\mathcal{H}_A)$ to represent the identity operator. Notation $\mathbb{Z}_n$ is used to represent all non negative integers less than n . We write $negl$ to denote any negligible function, which is a function f for every constant $c \in \mathbb{N}$ there exists $m \in \mathbb{N}$ such that for all $m > N$, $f(m) < m^{-c}$.

We use $F = \{f_{k,b} : X \rightarrow D_Y\}_{b \in \{0,1\}, k \in K_F}$ to represent an Extended Noisy Trapdoor Claw-free Family (ENTCF)[19], use $G = \{g_{k,b} : X \rightarrow D_Y\}_{b \in \{0,1\}, k \in K_G}$

to represent a Trapdoor Injective Function Family (INF)[19], the strict definition of ENTCF and INF can be seen in [19].

One-Way Secure State Generator (OWSG) is a state generator based on oneway-ness assumption, which can provide similar functions when one-way function assumption is not tenable.

Definition 1 (t -Copy OWSG) [10] OWSG is a set of quantum polynomial time (QPT) algorithms (**KeyGen**, **StateGen**, **Ver**) where

1. **KeyGen**(1^n). On input the security parameter n , output a classical key string $k \in \{0, 1\}^n$.

2. **StateGen**(k). On input key $k \in \{0, 1\}^n$, output an m -qubit quantum state $|\psi_k\rangle$.

3. **Ver**($k, |\psi\rangle$). On input key $k \in \{0, 1\}^n$ and m -qubit quantum state $|\psi\rangle$, output $\top$ or $\perp$. These algorithms satisfy the following properties.

1. Correctness. For every $n \in \mathcal{N}$,

$$\Pr(\top \leftarrow \mathbf{Ver}(k, |\psi_k\rangle)) > 1 - \text{negl},$$

where $k \leftarrow \mathbf{KeyGen}(1^n)$, $|\psi_k\rangle \leftarrow \mathbf{StateGen}(k)$.

2. t -Copy Security. For every quantum polynomial-sized adversary $\mathcal{A} = \{\mathcal{A}_n\}_{n \in \mathcal{N}}$ and $n \in \mathcal{N}$,

$$\Pr(\top \leftarrow \mathbf{Ver}(\mathcal{A}_n(|\psi_k\rangle^{\otimes t}), |\psi_k\rangle)) < \text{negl}$$

where $k \leftarrow \mathbf{KeyGen}(1^n)$, $|\psi_k\rangle \leftarrow \mathbf{StateGen}(k)$.

EFI pairs refer to two density operators with specified properties. Depending on the structure, the nature and form of EFI pairs will be different.

Definition 2 (EFI pairs) [12] We call $\xi = (\xi_{b,\lambda})$ a pair of EFI states if it satisfies the following criteria:

1. Efficient Generation. There exists a uniform QPT quantum algorithm A that on input $(1^\lambda, b)$ for some integer λ and $b \in \{0, 1\}$, outputs the mixed state $\xi_{b,\lambda}$.
2. Statistically Far. $\text{TD}(\xi_{0,\lambda}, \xi_{1,\lambda})$ as a function of λ is at least inverse polynomial.
3. Computational Indistinguishability. $(\xi_{0,\lambda})_\lambda$ is computationally indistinguishable to $(\xi_{1,\lambda})_\lambda$.

There is a specific construction of EFI pairs from OWSG.

Lemma 1 [18] Let $m = \frac{cn}{\log(n)}$ for some constant $c > 0$ where n is the key-length of the OWSG. An m -copy OWSG implies a k^* imbalanced EFI for some $k^*(\lambda) \in \text{poly}(\lambda)$.

A canonical (non-interactive) quantum bit commitment scheme provides a general architecture for quantum bit commitment. We can substitute pseudorandom state or EFI pars into it to obtain a specific quantum bit commitment protocol.

Definition 3 [11] A canonical (non-interactive) quantum bit commitment scheme is represented by an ensemble of polynomial-time uniformly generated quantum circuit pair $(Q_0(n), Q_1(n))_n$ as follows, where we drop the security parameter n to simplify the notation:

1. In the commit stage, to commit a bit $b \in \{0, 1\}$, the sender performs the quantum circuit Q_b on the quantum register pair (C, R) initialized in all $|0\rangle$'s state. Then the sender sends the commitment register C to the receiver, whose state at this moment is denoted by ρ_b .
2. In the subsequent (canonical) reveal stage, the sender announces the bit b , and sends the decommitment register R to the receiver. The receiver will first perform $Q_b^\dagger$ on the quantum register pair (C, R) and then measure each qubit of (C, R) in the computational basis, accepting if measurement outcomes are all 0's.

By the conclusion of Brakerski, Canetti and Qian[12], quantum bit commitment constructed from EFI pairs mentioned in definition 2 is computational hiding.

Lemma 2 [12] A commitment scheme satisfies computational hiding, if $\rho_{\lambda,0}$ is computationally indistinguishable to $\rho_{\lambda,1}$.

QKD can establish a shared key between two quantum participants. In this paper, we need a polynomial time quantum secure QKD protocol, but it does not depend on a specific structure. Therefore, we give a brief definition of QKD below. In practice, any QKD that meets the requirements can be applied to the conclusion of this paper.

Definition 4 (QKD) [13] QKD is a secure communication protocol that uses quantum - mechanical properties such as the uncertainty principle or quantum entanglement to distribute cryptographic keys between two parties (such as a sender and a receiver). It enables the generation of a shared, random, and secret key, which can then be used for secure encryption and decryption of messages, with the advantage that any eavesdropping attempt on the key - distribution process can be detected through the disturbance of the quantum states.

We introduce the definitions of VP and QVP below.

Definition 5 (Voting Protocol) VP includes two parties voters A_i and center C , and three stages: setting up voting, voting, and counting votes.

1. Setting Up. C inputs security parameter n and generates the voting information for voters.
 2. Voting Phase. A_i votes and outputs some corresponding information.
 3. Accounting. C verifies the information and counts voting results.
- These phases satisfy the following properties.
1. Correctness. Result from honest voters can be computed by C with the possibility close to $1 - \text{negl}$.
 2. Fairness. Each voter cannot obtain any message from others.
 3. Verifiability. Repeated votes are counted only once.

QVP is a quantum version of VP.

Definition 6 (Quantum Voting Protocol) QVP has two quantum participants, voters A_i and center C . It is a set of QPT algorithms (**StateGen**, **Vote**, **Ver**) where

1. **StateGen**. On input the security parameter n , output a pair n -qubit quantum states.
2. **Vote**. A_i votes and outputs a qubit quantum information.
3. **Ver**. On input one qubit from all voters, outputs classical information.

These algorithms satisfy the following properties.

1. Correctness. The probability that the voter's voting content is the same as the voting result recorded in the center is close to $1 - \text{negl}$.
2. Fairness. Each voter cannot obtain any message from others.
3. Verifiability. Repeated votes are counted only once.

3 Quantum Voting Protocol from EFI pairs

In this section, we will show how to construct our quantum voting protocol based on EFI pairs, and then analyse its properties.

3.1 The construction of our protocol

The protocol is an interactive process between quantum center C and N quantum voters A_i over quantum channel. The detail of our protocol is shown in figure 1.

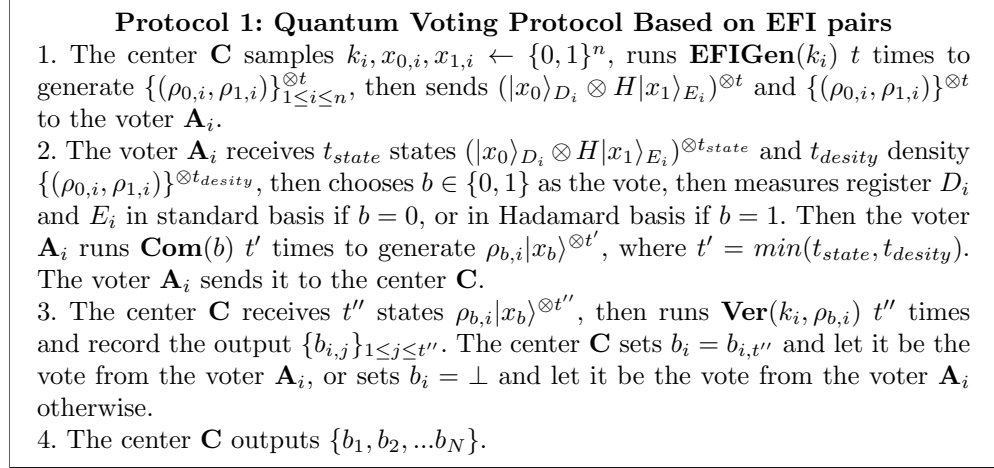


Fig. 1 The Detailed Process of Our Voting Protocol

3.2 Properties of our protocol

In this part, we analyze the interaction details of the protocol to demonstrate that it satisfies the properties required by the definition 6.

3.2.1 Correctness

Here we show that for an honest voter **A** who strictly follows protocol steps and performs no malicious actions, the center **C** can record the vote correctly with possibility close to 1. We formally describe it in the following theorem.

Theorem 1 (*Correctness*) *The output of center **C** is equal to the result of an honest voter **A** with the probability negligibly close to 1.*

Proof According to definition 1, the center **C** holding the key k_i can distinguish between $\rho_{0,i}$ and $\rho_{1,i}$, which means

$$\Pr(\mathbf{Ver}(k_i, \rho_{b,i}|x_b)) = b) > 1 - \text{negl},$$

where $k_i \leftarrow \mathbf{KeyGen}$, $\rho_{b,i} \leftarrow \mathbf{EFIGen}$. This means that for honest voter **A**, every result b_j obtained by the center should be equal to the voting result, which means

$$\Pr(b_i = b_j) > 1 - \text{negl},$$

where $i, j \in \mathcal{Z}_t$.

Since the protocol has only one interaction, it is easy to find that the correctness of the protocol relies entirely on the correct verification of the returned results. Therefore, for an honest voter **A**, the probability of the voting result being equal to the center record value should be close to 1, which means

$$|\Pr(b_i = b) - \Pr(\mathbf{Ver}(k_i, \rho_{b,i}|x_b)) = b)| < \text{negl}.$$

□

3.2.2 Fairness

In this part, we discuss the information leakage during the interaction process. The following theorem is established.

Theorem 2 (*Fairness*) *For each voter, the result of his vote is kept secret from semi-honest people, without any information leaked.*

Proof For the convenience of description, we consider that the center **C** only interacts with one voter **A**, and **A** is a semi-honest adversary in the interaction between the two sides. Note that the model can be trivially extended to the case of N voters. On this basis, we give **A** the strongest adversarial advantage, that is, we assume that **A** can obtain the quantum state transmitted by the center **C** and the voter **A** without being discovered and interfere with the interaction between the two sides, which means **A** receives $(|x_0\rangle \otimes H|x_1\rangle)^{\otimes t_A}$, $\{(\rho_{0,i}, \rho_{1,i})\}^{\otimes t_A}$ and $(\rho_{b,i}|x_b\rangle)^{\otimes t_A}$, where t_A is the number of copies of the states and densities received by **A**. Let's first prove the case of one cycle that the information about the vote b obtained by **A** should be negligible when $t_A = 1$.

By definition 2, it is hard to differentiate $\rho_{0,i}$ and $\rho_{1,i}$ in polynomial time for computational indistinguishability of EFI pairs, which means

$$\Pr(\mathcal{A}(\rho_{0,i}, \rho_{1,i}, \rho_{b,i}) = b) < \text{negl}.$$

Therefore, for $\mathcal{A}$, the probability of correctly distinguishing $\rho_{0,i}|x_0\rangle$ from $\rho_{1,i}|x_1\rangle$ is also negligible, which means

$$\Pr(\mathcal{A}(\rho_{0,i}|x_0\rangle, \rho_{1,i}|x_1\rangle, \rho_{b,i}|x_b\rangle) = b) < \text{negl}.$$

Since center **C** and voter **A** interact only once, and $\rho_{b,i}|x_b\rangle$ contains all the information about b , the inability to effectively distinguish $\rho_{0,i}|x_0\rangle$ from $\rho_{1,i}|x_1\rangle$ means that only having one intermediate state of interaction cannot give a non negligible advantage to obtain the value of b , which means

$$|\Pr(\mathcal{A}(|x_0\rangle \otimes H|x_1\rangle, \rho_{0,i}, \rho_{1,i}, \rho_{b,i}|x_b\rangle) = b) - \frac{1}{2}| < \text{negl}.$$

By the conclusion of [10], the fairness of our protocol can be reduced to BQP = QMA.

Now we prove the case of multiple cycles. When $t_{\mathcal{A}} > 1$, the establishment of theorem 2 can be reduced to the establishment of the t -copy security of EFI pairs. Specifically, from definition 1 and lemma 1, construction of EFI pairs from OWSG is also of t -copy security, that is to say, for a semi-honest adversary $\mathcal{A}$, the copy of the density $\rho_{1,i}$ and the state $|x_1\rangle$ will not bring a negligible advantage, which means

$$|\Pr(\mathcal{A}((|x_0\rangle \otimes H|x_1\rangle)^{\otimes t}, (\rho_{0,i}, \rho_{1,i})^{\otimes t}, (\rho_{b,i}|x_b\rangle)^{\otimes t}) = b) - \Pr(\mathcal{A}(|x_0\rangle \otimes H|x_1\rangle, \rho_{0,i}, \rho_{1,i}, \rho_{b,i}|x_b\rangle) = b)| < \text{negl}$$

Therefore, the establishment of multiple cycles can be reduced to one cycle. $\square$

3.2.3 Verifiability

Here we show that our voting protocol is verifiable, which can be described as the following theorem.

Theorem 3 (*Verifiability*) *The center **C** records one vote b_i for all voters $\mathbf{A}_i$.*

Proof The establishment of theorem 3 can be reduced to the third step of protocol 1, and the proof process is trivial. $\square$

3.3 Further analysis on Security

We have proved that our protocol satisfies correctness, fairness and verifiability, and will not disclose the voting content to any semi-honest adversaries. In this section, we analyze the security of our protocol in the presence of malicious adversary. Without losing generality, we consider the case where the number of voters is 1. The case of multiple voters can be reduced to the case of a single voter. The following theorem holds.

Theorem 4 *Let **C** be a quantum center interacting with a quantum voter **A** through a quantum channel, $\mathcal{A}$ is an arbitrary probabilistic polynomial time adversary, then*

$$|\Pr(\mathcal{A}(|x_0\rangle \otimes H|x_1\rangle, \rho_{0,i}, \rho_{1,i}, \rho_{b,i}|x_b\rangle) = b) - \Pr(\mathcal{A}(|\psi\rangle, \rho', \rho'', |\phi\rangle) = b)| < \text{negl} \quad (1)$$

where $|\psi\rangle, \rho', \rho'', |\phi\rangle$ are some arbitrary states and densities.

Proof The proof of theorem 4 is similar to theorem 2, therefore, We briefly explain the proof idea below. Due to the computational indistinguishability of EFI pairs, the probability of distinguishing between $\rho_{0,i}|x_0\rangle$ and $\rho_{1,i}|x_1\rangle$ is negligible, which implies the advantage for $\mathcal{A}$ provided $\rho_{0,i}$ and $\rho_{1,i}$ to obtain b is also negligible. Therefore, equation 1 holds. As concluded in [10], theorem 4 can be reduced to $\text{BQP} = \text{QMA}$. $\square$

There are two purposes to re-prove the advantage of $\mathcal{A}$ is negligible. One is to reduce the restrictions on $\mathcal{A}$. It is noted that compared with theorem 2, theorem 4 removes the assumption of semi-honesty of $\mathcal{A}$. Another purpose is to facilitate classifying the adversary's behaviors. In addition to stealing voters' votes, malicious adversaries in voting protocols also engage in acts such as forging votes and destroying votes. For example, if a malicious adversary intercepts the vote $\rho_b|x_b\rangle$ and tampers with $\rho_b|x_b\rangle$ to $\rho_{1-b}|x_{1-b}\rangle$, $\mathbf{C}$ cannot notice this change since $\mathbf{C}$ is unable to verify the identification of the voter and identify which vote is really from the voter. Based on the above analysis, we classify malicious adversaries into two categories, stealing adversaries and destroying adversaries. Stealing adversaries aim to obtain the vote b , while destroying adversaries intend to forge votes or prevent designated voters from participating in voting. Then we make the following claim.

Claim: Our quantum voting protocol is secure against stealing adversaries, yet it is unable to resist attacks from destroying adversaries.

4 Quantum Voting Protocol with QKD

To address the shortcomings of protocol 1, we use QKD to optimize it. Similar to the previous section, the optimized protocol is an interactive process between quantum center $\mathbf{C}$ and N quantum voters $\mathbf{A}_i$ over a quantum channel, where $N \in \mathbb{N}^+$ denotes the number of participating voters. The protocol details are shown in Figure 2.

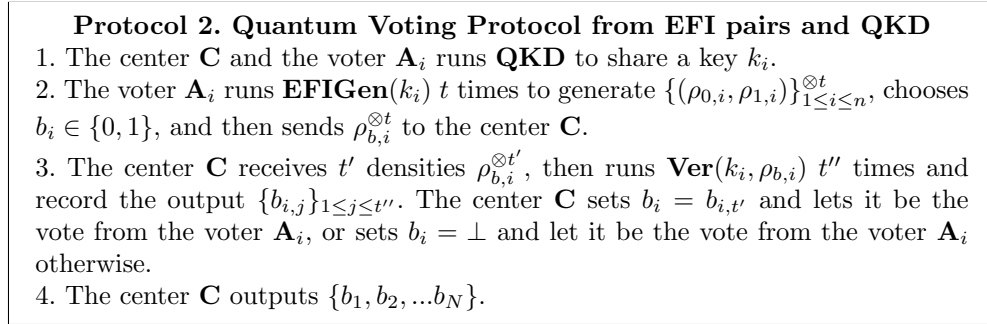


Fig. 2 The Detailed Process of Our Voting Protocol with QKD

Note that compared with protocol 1, protocol 2 establishes key k_i between the center and the voter, enabling both parties to verify the density operator received from each other. Moreover, we note that the objects transmitting voting information during the interaction remain EFI pairs, so the security analysis of protocol 2 is analogous

to that of protocol 1. In the following proofs, we will briefly explain these similarities, particularly focusing on their resistance to destroying adversaries.

Theorem 5 (Correctness) *The center $\mathbf{C}$ accepts the voting results from all honest voters with the probability negligibly close to 1.*

Proof The proof of theorem 5 is similar to that of theorem 1. As the key k_i is pre-shared, the voting information is directly encoded in the index of the density operator. By definition 1, theorem 5 holds. $\square$

Theorem 6 (Fairness) *Each voter's vote is kept secret from others.*

Proof For convenience of description, we consider a scenario where the center $\mathbf{C}$ interacts only with one voter $\mathbf{A}$, and $\mathcal{A}$ is an arbitrary probabilistic polynomial-time adversary. By the definition 2, we have

$$|\Pr(\mathcal{A}(\rho_{b,i}) = b) - \Pr(\mathcal{A}(\rho') = b)| < \text{negl},$$

where ρ' is an arbitrary density matrix, and

$$|\Pr(\mathcal{A}(\rho_{b,i}) = b) - \frac{1}{2}| < \text{negl}.$$

It is different from protocol 1. If $\mathcal{A}$ successfully forges the vote, it must be able to distinguish the density operator (note that the forged vote in protocol 1 only needs to apply the densities to the state) or steal the key k_i from the **QKD**. The advantages of these two methods should be stated separately, and their security can be respectively reduced to the security of **QKD** and the classical assumption i.e. $\text{BQP} = \text{QMA}$ [10]. $\square$

Theorem 7 (Verifiability) *Each vote can be counted only once.*

Proof The validity of theorem 7 can be reduced to the third step of protocol 2, and the proof process is straightforward. $\square$

5 Hybrid Quantum Voting Protocol

In this section, we propose our hybrid quantum voting protocol based on ENTCTF, and analyse the security of the protocol.

5.1 The Construction of Our Protocol

We use ENTCTF mentioned in section 2 to construct our protocol. Let λ be a security parameter. **ANGLE** is an algorithm on input two vector α and β , and outputs the angle ρ between them. Let X be $\{0, 1\}^n$, K_F be a finite set of keys. Let F be an ENTCTF, then there exists algorithms **GEN** _{F} and **INV** _{F} , where we denote the key generation algorithm as

$$(k, t_k)_F \leftarrow \mathbf{GEN}_F(1^\lambda),$$

and inversion algorithm as

$$\mathbf{INV}_F(t_k, b, y) = x.$$

where k is a key, t_k is a tag, and y, x are inputs and outputs respectively.

The algorithm **SAMP** can prepare the initial state on input a function f and an arbitrary original state $|\psi\rangle$, where f is an ENTCTF, and $|\psi\rangle$ can be written as

$$\alpha_0|0\rangle + \alpha_1|1\rangle,$$

where $\alpha_0, \alpha_1 \in \mathcal{C}, \alpha_0^2 + \alpha_1^2 = 1$. The effect of **SAMP** on $|\psi\rangle$ can be written as

$$\mathbf{SAMP}_f(|\psi\rangle) = \frac{1}{\sqrt{2|X|}} \sum_{b \in \{0,1\}} \sum_{x \in X} \alpha_b \sqrt{f_{k_i,b}} |b\rangle_C |x\rangle_D |f_{k_i,b}\rangle_E.$$

We use $|\psi\rangle_s$ to represent the state after applying **SAMP** to $|\psi\rangle$. Let C, D , and E represent the first, second, and third registers, respectively. We describe the measurement protocol in detail in figure 3.

5.2 Security Analysis

We first prove the correctness of protocol 3.

Theorem 8 *For honest voter and center, $\{p, 1-p\}$ and the voting transcript (VT) correspond one-to-one.*

Proof Without losing generality, we assume that the case of $vote = p$ and $vote = 1-p$ can be similarly obtained. Let v'_i be the measurement result of the center on the state in the specified basis and v_i be the expected result. So, for an honest center, the probability of $v'_i = v_i$ will not deviate from p . As a result, we have

$$|\mathbf{E}(v'_i - v_i) - vote| = |\mathbf{E}(v'_i - v_i) - p| = \text{negl}.$$

Therefore, if a voter sends p , the center must record 0. Similarly, if a voter sends $1-p$, the center must record 1. This implies that $\{p, 1-p\}$ and the VT have a one-to-one correspondence. $\square$

Note that our analysis of correctness covers only the second half of the interaction. This is because our verification process only covers the first half of the protocol, which is consistent with the reference[19], so it is not further expanded here. Additionally, when counting the interaction rounds, the n iterations in the protocol are recorded as a single round. This is because the iterations in the protocol do not necessitate chained computation. In other words, iterations are employed merely for descriptive convenience. Actually, these operations lack a sequential relationship and thus can be executed synchronously.

The next step is to verify fairness. The following theorem establishes.

Protocol 3. Hybrid Quantum Voting Protocol

1. The voter runs **GEN**_F to generate (k, t_k) , and then transmits $\{k\}$ to the center while keeping $\{t_k\}$ private.
2. The center runs the **SAMP** to obtain the state $|\psi_s\rangle$. Then, the voter measures the register E in the computational basis to get the measurement outcome $\{y\}$, and post measurement state $|\psi'\rangle$, then transmits $\{y\}$ to the verifier.
3. The voter receives $\{y\}$, runs **INV**_F (t_k, y) to obtain $\{\hat{b}, x_{\hat{b}, y}\}_{\hat{b} \in \{0,1\}}$.
4. The voter chooses $test \in \{0, 1\}$ randomly, and transmits it to the center.
5. If $test = 0$, the center measures remaining state in the computation basis, and then transmits the result (b', x'_b) to the voter. If $(b', x'_b) \neq (\hat{b}, x_{\hat{b}, y})$, the outputs of the voter is $\perp$, and the center aborts protocol.
6. If $test = 1$, the center performs Fourier basis measurement on the register D and obtains $d \in \mathcal{Z}_n$, then sends it to the voter.
7. The voter randomly chooses $bt \in \{0, 1\}$, $p \leftarrow (0, \frac{1}{2} - non - negl) \cup (\frac{1}{2} + non - negl, 1)$, and $v \in \{0, 1\}$ as the result, and computes α and β , where

$$\alpha = (\alpha_0, \alpha_1 e^{\frac{2i\pi}{n} d_i(x_{0,y} + x_{1,y})}),$$

and

$$\beta = (\frac{1}{\sqrt{2}}, (-1)^{v_i bt} \frac{1}{\sqrt{2}}).$$

Then, the voter runs **ANGLE** $\{\alpha, \beta\}$ and obtains the output ρ . The voter computes $\gamma = |\arccos p - \rho|$ and sends (γ, bt) to the center.

8. The center does a $R_z(\gamma)$ on register C , and does request basis measurement on the state. $bt = 0$ represents $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$, and $bt = 1$ represents $|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$. Then the center obtains the measurement outcome v' .
9. The voter receives v' from the prover, and then fixes $test = 1$ and repeats step 1 to step 9 $2n$ times.
10. The center sends $V' = \{v'_1, \dots, v'_n\}$ to the voter. If $|\mathbf{E}(v'_i = v_i) - p| > negl$, the voter will output $\perp$ and aborts the protocol.
11. Then the voter sends $V = \{v_{n+1}, \dots, v_{2n}\}$ and $vote$ to the center, where $vote \in \{p, 1-p\}$. The center calculates $s = |\mathbf{E}(v'_i = v_i) - vote|$, and records VT as the vote from the voter, where $VT = 0$ if $s = negl$, $VT = 1$ if $s + 2vote - 1 = negl$, $VT = \perp$ and abort otherwise.

Fig. 3 The Detailed Process of Our Voting Protocol

Theorem 9 *Let A be the voter and C be the center. Then, for any quantum polynomial-time adversary ADV without k , the probability of obtaining or effectively tampering with voting content is negligible, which means*

$$\Pr(ADV(y, d, \gamma, bt, vote, v'_1, \dots, v'_n, v_{n+1}, \dots, v_{2n}) = VT) < negl.$$

Proof From the construction of $|\psi_s\rangle$, the measurement outcome y is equivalent to any element in the range of ENTCF. Therefore, according to the assumption of ENTCF, y alone cannot

generate $|\psi'\rangle$, which means

$$\Pr(ADV(y) = |\psi'\rangle) < \text{negl}. \quad (2)$$

Similarly, ADV cannot effectively tamper with y , as such tampering will be detected in step 5.

In addition, according to the assumption of ENTCTF, ADV cannot generate the state $|\psi''\rangle$ contained in register C . Moreover γ contains a random value selected by the voter. As a result, the probability of ADV obtaining $|\psi''\rangle$ is negligible, which means

$$\Pr(ADV(y, d, \gamma) = |\psi''\rangle) < \text{negl}. \quad (3)$$

Finally, note that the voting content is embedded in the comparison of two sets of probability distributions. The protocol's structure ensures that the transmitted information contains only one set of these distributions. Thus, half of the distribution finally obtained by ADV comes from the first half and the other half from the second half. This implies that ADV cannot perform an effective comparison, which means

$$\Pr(ADV(\text{vote}, v'_1, \dots, v'_n, v_{n+1}, \dots, v_{2n}) = VT) < \text{negl}.$$

Combined with equation 2 and 3, we conclude that the adversary can neither obtain voting information from the interaction nor alter the interaction process to tamper with the voting information in a directional manner. In summary, we have

$$|\Pr(ADV(y, d, \gamma, bt, \text{vote}, v'_1, \dots, v'_n, v_{n+1}, \dots, v_{2n}) = VT) - \Pr(ADV(\lambda) = VT)| < \text{negl}.$$

□

It is worth mentioning that we restrict the adversary from obtaining k , because if the adversary obtains k , it can masquerade as the center to obtain the voting content. However, due to the computational power gap between the classical party and the quantum parties, it is difficult to incorporate anti-forgery mechanisms in the protocol without information gap. As in section 4, we can have both parties implement a key distribution protocol in advance to avoid this restriction. The structural embedding of this part is similar to protocol 2, so we will not discuss it separately.

The last step is to prove the verifiability.

Theorem 10 *The center records only one vote for every voter.*

Proof Note that regardless of the number of states sent, the center will ultimately obtain only one probability distribution, and the verification information corresponding to the vote is also unique. Therefore, the center will not record additional votes when interacting with the same voter. □

6 Conclusion

In this paper, we propose the first quantum voting protocol based on mathematical difficulty assumptions and the first hybrid quantum voting protocol. In the protocol design process, we replace the entanglement of states in traditional quantum voting protocol structures with the transmission of quantum states, and replace quantum operations with EFI pairs. Additionally, based on ENTCTF, we substitute classical

counterparts for the voter and channel in the quantum voting protocol. This substitution can effectively avoid the dependence of properties on devices and channels, enabling completion under more extreme conditions. Moreover, our protocol can replace plugins, making it adaptable to a wider range of scenarios. It is worth mentioning that using EFI pairs to replace entangled states can reduce dependence on the environment and equipment. At the same time, the security of EFI pairs can be strictly reduced to the classical assumption, i.e., $BQP = QMA$. This allows our protocol to have both the convenience of classical algorithms and the efficiency of quantum algorithms.

However, there are still some improvements needed in our work. First, compared to existing protocols, our protocol requires more storage space for quantum states, due to EFI pairs having certain requirements on the length of qubits. If the length is too short, the definition requirements of EFI pairs are not met. Second, our quantum state needs to transmit more quantum bits, as our protocol needs to authenticate the transmitted state, and this process consumes protocol resources. Finally, without QKD, our protocol is difficult to resist malicious adversaries' tampering with the final voting results, which is a common security requirement in quantum voting protocols. But using QKD for every voter is wasteful, especially in the case of a single vote.

Therefore, under the premise of ensuring security, compressing the length of qubits is a worthwhile optimization direction. Additionally, it is also worth considering how to design protocols to reduce the number of quantum states transmitted and the number of interactive rounds. Furthermore, both participants in our protocol are fully quantum, which makes it difficult to apply in practical scenarios, especially when the development of quantum computers is not yet complete. Therefore, it is very meaningful to consider the limitation of participants' ability, for example, in scenarios where a participant can be semi-quantum or mixed quantum protocol. In addition, classifying the property as $BQP=QMA$ does not necessarily result in a completely compact protocol. Therefore, exploring the compactness of the protocol and providing the minimum security assumption of QVP remains an urgent problem.

Declarations

- Data availability Not applicable
- Competing interests The authors declare that they have no competing interests.
- Funding The research was funded by the National Key Research and Development Program of China (No. 2021YFA1000600), the National Natural Science Foundation of China (No. 12441104) and the Science and Technology Development Fund, Macau SAR (File no. 0084/2022/A).
- Author contribution Tingyu Ge complete the design of the protocol and the analysis and proof of its security. All authors read and approved the final manuscript.
- Acknowledgements Not applicable.

References

- [1] Shor P W. Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer[J]. SIAM review, 1999, 41(2): 303-332.

- [2] Li, Yannan, et al. A blockchain-based self-tallying voting protocol in decentralized IoT[J]. *IEEE Transactions on Dependable and Secure Computing* 19.1 (2020): 119-130.
- [3] Cortier, Véronique, David Galindo, and Mathieu Turuani. A formal analysis of the neuchâtel e-voting protocol[J]. *2018 IEEE European Symposium on Security and Privacy*. IEEE, 2018 : 430-442.
- [4] Vaccaro J A, Spring J, Cheffles A. Quantum protocols for anonymous voting and surveying[J]. *Physical Review A, Atomic, Molecular, and Optical Physics*, 2007, 75(1): 012333.
- [5] Jiang L, He G, Nie D, et al. Quantum anonymous voting for continuous variables[J]. *Physical Review A—Atomic, Molecular, and Optical Physics*, 2012, 85(4): 042309.
- [6] Wang Q, Yu C, Gao F, et al. Self-tallying quantum anonymous voting[J]. *Physical Review A*, 2016, 94(2): 022333.
- [7] Sakhuja S, Balakrishnan S. Quantum-Enhanced Secure Approval Voting Protocol. *arXiv preprint arXiv:2406.19730*, 2024.
- [8] Ji Z, Liu Y K, Song F. Pseudorandom quantum states[C]//*Advances in Cryptology—CRYPTO 2018: 38th Annual International Cryptology Conference*, Santa Barbara, CA, USA, August 19–23, 2018, Proceedings, Part III 38. Springer International Publishing, 2018: 126-152.
- [9] Kretschmer W. Quantum Pseudorandomness and Classical Complexity[C]//*16th Conference on the Theory of Quantum Computation, Communication and Cryptography*. 2021.
- [10] Morimae T, Yamakawa T. Quantum commitments and signatures without one-way functions[C]//*Annual International Cryptology Conference*. Cham: Springer Nature Switzerland, 2022: 269-295.
- [11] Yan J. General properties of quantum bit commitments[C]//*International Conference on the Theory and Application of Cryptology and Information Security*. Cham: Springer Nature Switzerland, 2022: 628-657.
- [12] Brakerski Z, Canetti R, Qian L. On the Computational Hardness Needed for Quantum Cryptography[C]//*14th Innovations in Theoretical Computer Science Conference (ITCS 2023)*. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2023.
- [13] Cryptography Q. Public key distribution and coin tossing[C]//*Proc. of IEEE Int. Conf. on Comput. Sys. and Sign. Proces.*, Bangalore, India. 1984: 175-179.

- [14] Santos M B, Mateus P, Pinto A N. Quantum oblivious transfer: a short review[J]. *Entropy*, 2022, 24(7): 945.
- [15] Bennett C H, Brassard G, Crépeau C, et al. Practical quantum oblivious transfer[C]//Annual international cryptology conference. Berlin, Heidelberg: Springer Berlin Heidelberg, 1991: 351-366.
- [16] Khurana D, Tomer K. Commitments from quantum one-wayness[C]//Proceedings of the 56th Annual ACM Symposium on Theory of Computing. 2024: 968-978.
- [17] Morimae T, Yamakawa T. One-Wayness in Quantum Cryptography[C]//19th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2024). Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2024.
- [18] Batra, Rishabh, and Rahul Jain. Commitments are equivalent to one-way state generators[J]. *arxiv preprint arxiv:2404.03220*, 2024.
- [19] Mahadev U. Classical verification of quantum computations[C]//2018 IEEE 59th Annual Symposium on Foundations of Computer Science (FOCS). IEEE, 2018: 259-267.